HCIA-AI V4.0 Certification Practice Exam • Chapter 2

Module 2: Machine Learning Overview (100 Questions & Answer Key)

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Question 1

TRUE OR FALSE • 16 Points

L1 regularization is commonly referred to as weight decay in machine learning optimization.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

In a random search for hyperparameters, each setting item is sampled from possible parameter values independently based on a specified probability distribution.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

Assuming a dataset contains the areas and historical sale prices of 21,613 houses, predicting the price of an unseen house based on its area is a typical regression task.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

In machine learning regression tasks, Mean Squared Error (MSE) is one of the most commonly used loss functions.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

Gradient descent is the only optimization method available for training machine learning algorithms.

A. True
B. False

Page Answers: **Q1**: False • **Q2**: True • **Q3**: True • **Q4**: True • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

If the hyperparameter search space is large and high-dimensional, random search is usually more computationally efficient than exhaustive grid search.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

In supervised learning, the training dataset must contain only unlabelled inputs without any ground-truth target labels.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

K-means clustering requires pre-labeled training samples to establish class boundaries.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

High variance in a machine learning model typically indicates that the model is suffering from underfitting on the training dataset.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

L1 regularization tends to produce sparse weight vectors by driving uninformative feature weights strictly to zero, effectively performing feature selection.

A. True
B. False

Page Answers: **Q6**: True • **Q7**: False • **Q8**: False • **Q9**: False • **Q10**: True

Question 11

TRUE OR FALSE • 16 Points

The k-Nearest Neighbors (KNN) algorithm is an instance-based lazy learning method that does not construct an explicit internal model during training.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

A smaller value of k in KNN makes the decision boundary smoother and reduces sensitivity to noisy outlier points.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

Naive Bayes assumes that all feature attributes are conditionally independent of each other given the class label.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

Support Vector Machines (SVM) can solve linearly non-separable problems in original feature spaces by using kernel functions to map data into higher-dimensional spaces.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

In Decision Trees, the ID3 algorithm uses the Gini Impurity criterion to select the optimal splitting attribute.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: False • **Q13**: True • **Q14**: True • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

Random Forest builds multiple independent decision trees using bootstrap data samples and random feature subsets to reduce prediction variance.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

In Boosting algorithms (such as AdaBoost and GBDT), base learners are trained in parallel completely independently of one another.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

Standardization (Z-score normalization) scales feature values strictly between 0 and 1, forcing all values to be non-negative.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

In Reinforcement Learning, the learning agent improves its policy by receiving immediate scalar reward signals from the environment based on its chosen actions.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

One-Hot Encoding is primarily used to convert continuous floating-point target variables into discrete categories for regression tasks.

A. True
B. False

Page Answers: **Q16**: True • **Q17**: False • **Q18**: False • **Q19**: True • **Q20**: False

Question 21

TRUE OR FALSE • 16 Points

The Area Under the ROC Curve (AUC) is equal to 1.0 for a perfect binary classification model.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Stochastic Gradient Descent (SGD) updates model parameters using the entire training dataset at the same time in each single iteration step.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

In cross-validation, the test dataset is merged into the training set to prevent the model from underfitting.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Linear regression models are naturally immune to multicollinearity among input features.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

In Logistic Regression, the Sigmoid function maps any real-valued number into a probability range between 0 and 1.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: False • **Q24**: False • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Without considering any regularization terms, the support vectors of a Support Vector Machine (SVM) are composed of which data points?

- **A.** The points closest to the separating hyperplane
- **B.** Points of a certain arbitrary class
- **C.** The points farthest from the separating hyperplane
- **D.** Points on the separating hyperplane

Question 27

SINGLE-ANSWER QUESTION • 16 Points

A computer uses labeled images to learn and distinguish between cats and dogs. What type of machine learning does this scenario represent?

- **A.** Self-supervised clustering
- **B.** Supervised learning
- **C.** Reinforcement learning
- **D.** Unsupervised learning

Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements is TRUE regarding classification and regression in supervised learning?

- **A.** Classification predicts continuous numerical values, whereas regression predicts discrete categories
- **B.** Both classification and regression predict discrete categorical labels
- **C.** Both classification and regression predict continuous numerical values
- **D.** Classification predicts discrete categorical labels, whereas regression predicts continuous numerical values

Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements about Support Vector Machines (SVM) is FALSE?

- **A.** Non-linear SVM uses kernel functions to project data into higher dimensions
- **B.** The support vectors determine the location and orientation of the optimal hyperplane
- **C.** SVM can only be used for unsupervised clustering without class labels
- **D.** The objective of linear SVM is to find a hyperplane that maximizes the margin between classes

Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an unsupervised learning algorithm?

- **A.** Principal Component Analysis
- **B.** Decision Tree
- **C.** K-Means clustering
- **D.** Hierarchical clustering

Page Answers: **Q26**: A • **Q27**: B • **Q28**: D • **Q29**: C • **Q30**: B

Question 31

SINGLE-ANSWER QUESTION • 16 Points

In Tom Mitchell's well-known definition of machine learning, an algorithm learns from Experience (E) with respect to some Task (T) and Performance measure (P) if its performance on T, as measured by P, improves with E. What does 'E' represent in this context?

- **A.** Historical data and interactions
- **B.** CPU energy consumption
- **C.** Computational execution speed
- **D.** Validation loss variance

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which of the following metrics is best suited for evaluating a linear regression model?

- **A.** F1-Score
- **B.** Cross-Entropy Loss
- **C.** Precision and Recall
- **D.** Mean Squared Error

Question 33

SINGLE-ANSWER QUESTION • 16 Points

Which of the following feature selection approaches evaluates feature subsets by training an actual predictive model and measuring its validation score?

- **A.** Random projection method
- **B.** Filter method
- **C.** Wrapper method
- **D.** Constant variance threshold

Question 34

SINGLE-ANSWER QUESTION • 16 Points

When an overly complex machine learning model achieves 99.9% accuracy on the training set but only 52% accuracy on the test set, what problem is it experiencing?

- **A.** High Bias
- **B.** Vanishing Gradient
- **C.** Overfitting
- **D.** Underfitting

Question 35

SINGLE-ANSWER QUESTION • 16 Points

Which regularization technique adds the sum of absolute values of the coefficients (|w|) as a penalty term to the loss function?

- **A.** L1 Regularization
- **B.** L2 Regularization
- **C.** Dropout
- **D.** Batch Normalization

Page Answers: **Q31**: A • **Q32**: D • **Q33**: C • **Q34**: C • **Q35**: A

Question 36

SINGLE-ANSWER QUESTION • 16 Points

Which regularization technique adds the sum of squared values of the coefficients (w^2) as a penalty term to the loss function?

- **A.** Softmax Function
- **B.** L1 Regularization
- **C.** L2 Regularization
- **D.** Early Stopping

Question 37

SINGLE-ANSWER QUESTION • 16 Points

Which splitting criterion is used by the CART (Classification and Regression Trees) algorithm for classification trees?

- **A.** Information Gain Ratio
- **B.** Information Gain
- **C.** Mean Absolute Error
- **D.** Gini Impurity

Question 38

SINGLE-ANSWER QUESTION • 16 Points

Which splitting criterion is used by the C4.5 algorithm to reduce ID3's bias toward attributes with large numbers of distinct values?

- **A.** Mean Squared Error
- **B.** Information Gain Ratio
- **C.** Gini Index
- **D.** Information Gain

Question 39

SINGLE-ANSWER QUESTION • 16 Points

In a binary classification confusion matrix, what does 'Recall' (Sensitivity) measure?

- **A.** TP / (TP + FN) - Proportion of actual positives correctly identified
- **B.** TP / (TP + FP) - Proportion of predicted positives that are truly positive
- **C.** (TP + TN) / (TP + TN + FP + FN) - Overall proportion of correct predictions
- **D.** TN / (TN + FP) - Proportion of actual negatives correctly identified

Question 40

SINGLE-ANSWER QUESTION • 16 Points

In a binary classification confusion matrix, what does 'Precision' measure?

- **A.** (TP + TN) / (TP + TN + FP + FN) - Overall proportion of correct predictions
- **B.** TP / (TP + FN) - Proportion of actual positives correctly identified
- **C.** TN / (TN + FP) - Proportion of actual negatives correctly identified
- **D.** TP / (TP + FP) - Proportion of predicted positives that are truly positive

Page Answers: **Q36**: C • **Q37**: D • **Q38**: B • **Q39**: A • **Q40**: D

Question 41

SINGLE-ANSWER QUESTION • 16 Points

What is the harmonic mean of Precision and Recall commonly known as?

- **A.** R-Squared
- **B.** Accuracy
- **C.** F1-Score
- **D.** ROC Curve

Question 42

SINGLE-ANSWER QUESTION • 16 Points

In the k-nearest neighbors (k-NN) classification algorithm, how is the class of an unlabeled sample typically predicted?

- **A.** By calculating the gradient of the loss function
- **B.** By constructing a maximum-margin separating hyperplane
- **C.** By majority voting among the k nearest neighbors
- **D.** By minimizing within-cluster variance

Question 43

SINGLE-ANSWER QUESTION • 16 Points

In the k-nearest neighbors (k-NN) algorithm, what is the effect of selecting a very small value for k?

- **A.** The decision boundary becomes completely linear
- **B.** The model has a higher probability of underfitting
- **C.** The model has a higher probability of overfitting due to noise
- **D.** The model requires zero distance calculations

Question 44

SINGLE-ANSWER QUESTION • 16 Points

In K-Means clustering, what does 'K' represent?

- **A.** The number of clusters to be partitioned
- **B.** The number of nearest neighbors for voting
- **C.** The dimensionality of input features
- **D.** The number of iterations until convergence

Question 45

SINGLE-ANSWER QUESTION • 16 Points

How are cluster centroids updated in standard K-Means clustering during each iterative step?

- **A.** By projecting support vector coordinates onto the hyper-plane boundary
- **B.** By computing the arithmetic mean of all data points currently assigned to that cluster
- **C.** By re-initializing coordinates randomly from a uniform Gaussian probability distribution
- **D.** By selecting the extreme boundary outlier with the maximum Euclidean distance

Page Answers: **Q41**: C • **Q42**: C • **Q43**: C • **Q44**: A • **Q45**: B

Question 46

SINGLE-ANSWER QUESTION • 16 Points

What is the primary operational mechanism of Bagging (Bootstrap Aggregating) in ensemble learning?

- **A.** Training base models sequentially where each model fits the residuals of previous learners
- **B.** Pruning deep decision branches to enforce low variance across individual shallow trees
- **C.** Discarding half of the training feature dimensions randomly during each gradient update
- **D.** Training multiple base models in parallel on bootstrapped subsets and aggregating their outputs

Question 47

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a classic Boosting algorithm?

- **A.** AdaBoost
- **B.** Principal Component Analysis
- **C.** Random Forest
- **D.** K-Means

Question 48

SINGLE-ANSWER QUESTION • 16 Points

In Naive Bayes classification, which principle provides the foundation for calculating posterior class probabilities?

- **A.** Bayes' Theorem
- **B.** Gradient Descent Optimization
- **C.** Decision Tree Pruning
- **D.** K-Means Centroid Allocation

Question 49

SINGLE-ANSWER QUESTION • 16 Points

What fundamental assumption does the Naive Bayes classification algorithm make regarding feature attributes?

- **A.** All features follow a uniform spherical Gaussian distribution
- **B.** All features are linearly dependent on the class label
- **C.** All features are conditionally independent of each other given the class label
- **D.** Features must be continuously normalized between 0 and 1

Question 50

SINGLE-ANSWER QUESTION • 16 Points

In gradient descent, what happens if the learning rate is configured to be excessively large?

- **A.** The loss value may oscillate and fail to converge, or even diverge completely
- **B.** The model parameters freeze permanently and cease executing mathematical updates
- **C.** The algorithm converges instantaneously to the exact global loss minimum on epoch 1
- **D.** The GPU physical RAM allocation drops to zero and clears all dataset tensors

Page Answers: **Q46**: D • **Q47**: A • **Q48**: A • **Q49**: C • **Q50**: A

Question 51

SINGLE-ANSWER QUESTION • 16 Points

Which gradient descent variant calculates gradients and updates parameters using a small subset (mini-batch) of training samples at each step?

- **A.** Full Grid Search
- **B.** Batch Gradient Descent
- **C.** Stochastic Gradient Descent
- **D.** Mini-batch Gradient Descent

Question 52

SINGLE-ANSWER QUESTION • 16 Points

In K-Fold Cross-Validation, if K = 5, what percentage of the dataset is used for validation in each fold?

- **A.** 80%
- **B.** 50%
- **C.** 10%
- **D.** 20%

Question 53

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the validation set in the machine learning workflow?

- **A.** Storing redundant duplicate records
- **B.** Hyperparameter tuning and model architecture selection
- **C.** Final unbiased assessment before production deployment
- **D.** Updating the internal weights via backpropagation

Question 54

SINGLE-ANSWER QUESTION • 16 Points

In machine learning feature engineering, what is the primary purpose of normalizing and standardizing features?

- **A.** To ensure that different input variables of a model fall into the same value range
- **B.** To convert categorical labels into one-hot encoded binary vectors
- **C.** To artificially increase training data by generating synthetic noise
- **D.** To eliminate negative weight coefficients across neural connections

Question 55

SINGLE-ANSWER QUESTION • 16 Points

What type of data problem occurs when an age column contains values like -15 or 999?

- **A.** Underfitting
- **B.** Outlier data
- **C.** Generalization
- **D.** Regularization

Page Answers: **Q51**: D • **Q52**: D • **Q53**: B • **Q54**: A • **Q55**: B

Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an effective technique to prevent decision trees from overfitting?

- **A.** Feature scaling
- **B.** Increasing tree depth
- **C.** Removing stopping criteria
- **D.** Tree pruning

Question 57

SINGLE-ANSWER QUESTION • 16 Points

What is the function of the Softmax layer in multi-class classification models?

- **A.** Normalizing raw model logits into a valid probability distribution summing to 1
- **B.** Computing pairwise Euclidean distance matrices between all feature sample vectors
- **C.** Eliminating all negative weight coefficients across dense hidden layer parameters
- **D.** Projecting multidimensional continuous inputs onto a binary step threshold output

Question 58

SINGLE-ANSWER QUESTION • 16 Points

In Support Vector Machines (SVM), what is the primary objective of constructing an optimal hyperplane?

- **A.** To minimize the number of input features used during training
- **B.** To maximize the margin (gap) between the two categories in the feature space
- **C.** To compute pairwise Euclidean distances to all training points
- **D.** To eliminate outliers by pruning decision tree leaf nodes

Question 59

SINGLE-ANSWER QUESTION • 16 Points

In Support Vector Machines (SVM), which kernel function is most frequently used to handle non-linearly separable data?

- **A.** Linear kernel
- **B.** Gaussian kernel
- **C.** Step kernel
- **D.** Constant kernel

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

In the machine learning workflow, why is Feature Selection necessary? (Multiple Choice)

- **A.** Simplifies models for easier interpretation
- **B.** Shortens model training time
- **C.** Eliminates the need for any training datasets or labels
- **D.** Avoids the curse of dimensionality and improves generalization

Page Answers: **Q56**: D • **Q57**: A • **Q58**: B • **Q59**: B • **Q60**: ABD

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

In linear regression models expressed as y = w^T x + b, which of the following statements are correct?

- **A.** 'w' represents the weight vector determining the slope of each feature
- **B.** 'b' represents the regularization hyperparameter controlling L1 sparsity penalties
- **C.** 'b' represents the bias term allowing the hyperplane to shift away from the origin
- **D.** The parameter vector 'w' must always be strictly perpendicular to the input feature vector 'x'

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following ensemble learning algorithms are based on sequential boosting and iterative residual fitting?

- **A.** AdaBoost
- **B.** Gradient Boosted Decision Trees
- **C.** Random Forest
- **D.** Extra Trees

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical application scenarios of Reinforcement Learning?

- **A.** Customer segmentation based on purchasing history
- **B.** Autonomous vehicle driving and traffic decisions
- **C.** Playing complex strategic games such as Chess and Go
- **D.** Industrial robot path planning and motion control

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

When evaluating classification models on imbalanced datasets (e.g., fraud detection where positive cases are 0.1%), which of the following evaluation strategies are appropriate?

- **A.** Precision, Recall, and F1-Score
- **B.** Relying only on overall Accuracy without inspecting the Confusion Matrix
- **C.** Area Under the Precision-Recall Curve or ROC-AUC
- **D.** Evaluating Mean Absolute Error on categorical predictions

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Semi-Supervised Learning are correct?

- **A.** It uses a small amount of labeled data combined with a large pool of unlabeled data
- **B.** It helps reduce the high cost and labor required for detailed manual data annotation
- **C.** It completely eliminates the need for any initial ground-truth supervision or seed labels
- **D.** It assumes that neighboring data points in feature space tend to share similar task labels

Page Answers: **Q61**: AC • **Q62**: AB • **Q63**: BCD • **Q64**: AC • **Q65**: ABD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following evaluation metrics are specifically designed for classification models rather than continuous regression?

- **A.** F1-Score
- **B.** Mean Squared Error
- **C.** Area Under the ROC Curve
- **D.** Mean Absolute Percentage Error

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hyperparameters may be defined and tuned during model training?

- **A.** Model weights and biases
- **B.** Number of training iterations
- **C.** Batch size
- **D.** Learning rate

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about model parameters and hyperparameters are true?

- **A.** Weights and biases in neural networks are model parameters
- **B.** Hyperparameters are set manually by developers prior to model training
- **C.** Learning rate and tree maximum depth are hyperparameters
- **D.** Model parameters are learned and updated automatically by the algorithm from training data

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about datasets in machine learning are true?

- **A.** The test set should be used to update gradient weights during training
- **B.** The training set is used to optimize model parameters
- **C.** The test set is used to evaluate final model generalization performance
- **D.** The validation set is used for hyperparameter tuning and model selection

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following tasks belong to the data preprocessing stage in machine learning?

- **A.** Data deduplication
- **B.** Missing value imputation
- **C.** Feature normalization and standardization
- **D.** Outlier detection and handling

Page Answers: **Q66**: AC • **Q67**: BCD • **Q68**: ABCD • **Q69**: BCD • **Q70**: ABCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following methods can help prevent model overfitting?

A. Applying cross-validation during hyperparameter selection
B. Expanding the training dataset size or applying data augmentation
C. Adding L1 or L2 regularization terms to the loss function
D. Adding millions of redundant parameters to a tiny dataset

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are unsupervised machine learning algorithms?

A. Multiple Linear Regression
B. K-Means Clustering
C. Principal Component Analysis
D. Support Vector Machines

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

When evaluating continuous regression models in the presence of extreme anomalous data points (outliers), which of the following statements regarding MSE, RMSE, and MAE are correct?

A. Mean Squared Error (MSE) squares prediction errors, causing large outliers to heavily distort the score
B. Root Mean Squared Error (RMSE) preserves the original unit scale of the target variable, making it more intuitive than MSE
C. Mean Squared Error (MSE) is completely immune to extreme outliers regardless of error magnitude
D. Mean Absolute Error (MAE) avoids squaring errors and is therefore the most robust regression metric when outliers exist

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following ensemble learning algorithms are based on Boosting principles?

A. AdaBoost
B. Random Forest
C. XGBoost
D. Gradient Boosting Decision Tree fitting negative gradients

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about K-Nearest Neighbors (KNN) are true?

A. KNN makes predictions based on the class labels of the k closest training examples
B. Feature scaling is essential for KNN to prevent high-magnitude features from dominating distance calculations
C. KNN requires weeks of intensive offline gradient descent training before it can make predictions
D. KNN is sensitive to the choice of distance metric

Page Answers: **Q71**: ABC • **Q72**: BC • **Q73**: ABD • **Q74**: ACD • **Q75**: ABD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Support Vector Machines (SVM) are correct?

A. Data points located on the margin boundaries are called support vectors

B. The optimal hyperplane maximizes the geometric margin between classes

C. Kernel functions allow SVM to construct non-linear decision boundaries

D. Standard linear SVM requires quadratic memory that decreases as feature dimensions grow

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are commonly used kernel functions in Support Vector Machines?

A. Linear Kernel

B. Sigmoid Kernel

C. Radial Basis Function

D. Polynomial Kernel

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following impurity and splitting criteria are commonly used in Decision Tree algorithms (ID3 and CART)?

A. Information Gain

B. Silhouette Separation Coefficient

C. Gini Impurity Index

D. Cosine Dissimilarity Metric

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following metrics are used to evaluate the performance of continuous regression models?

A. Mean Squared Error

B. Mean Absolute Error

C. Classification Accuracy

D. Recall or Sensitivity

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Gradient Descent variants are true?

A. Stochastic Gradient Descent updates parameters using a single random sample per step, making it fast but causing fluctuating loss curves

B. Mini-batch Gradient Descent balances computational stability and speed by using small subsets of data

C. Batch Gradient Descent uses all training samples to calculate gradient, guaranteeing smooth convergence but being computationally slow on massive data

D. All gradient descent methods update parameters without using derivatives

Page Answers: **Q76**: ABC • **Q77**: ABCD • **Q78**: AC • **Q79**: AB • **Q80**: ABC

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are automated search strategies commonly used for model hyperparameter tuning?

- **A.** Grid Search
- **B.** Random Search
- **C.** Constant Value Imputation
- **D.** One-Hot Category Encoding

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core components in a Reinforcement Learning system?

- **A.** State and Action
- **B.** Agent
- **C.** Reward signal
- **D.** Environment

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common statistical imputation strategies for handling missing values in continuous numeric features?

- **A.** Mean Value Imputation
- **B.** Random Label Flipping
- **C.** Median Value Imputation
- **D.** Softmax Probability Normalization

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are benefits of Feature Scaling (Normalization / Standardization)?

- **A.** Improves numerical stability of optimization algorithms
- **B.** Speeds up gradient descent convergence
- **C.** Prevents features with large scales from dominating distance-based models
- **D.** Guarantees 100% test accuracy on any arbitrary model

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Random Forest are correct?

- **A.** It requires base trees to be trained strictly in serial order
- **B.** It is generally more resistant to overfitting than an unpruned single decision tree
- **C.** It uses bootstrap aggregation to generate diverse training subsets
- **D.** It selects a random subset of features at each split node

Page Answers: **Q81**: AB • **Q82**: ABCD • **Q83**: AC • **Q84**: ABC • **Q85**: BCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are geometric spatial distance metrics commonly used in KNN and K-Means?

- **A.** Euclidean Distance
- **B.** Manhattan Distance
- **C.** Categorical Cross-Entropy Loss
- **D.** Margin Hinge Loss Function

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about K-Means clustering are true?

- **A.** The results can vary depending on the initial centroid positions
- **B.** It is sensitive to extreme outlier points
- **C.** It is a supervised classification technique that requires pre-annotated ground-truth labels
- **D.** It requires specifying the number of clusters K in advance

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common methods of feature selection in machine learning?

- **A.** Filter method
- **B.** Wrapper method
- **C.** Embedded method
- **D.** Gradient descent optimization

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Logistic Regression are true?

- **A.** Despite its name, Logistic Regression is primarily used for classification problems
- **B.** It optimizes model parameters using maximum likelihood estimation
- **C.** It uses the Sigmoid function to output probability values between 0 and 1
- **D.** It can only be used when there are at least 1,000 output classes

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

What four basic outcomes are recorded in a binary confusion matrix?

- **A.** True Positive
- **B.** False Negative
- **C.** False Positive
- **D.** True Negative

Page Answers: **Q86**: AB • **Q87**: ABD • **Q88**: ABC • **Q89**: ABC • **Q90**: ABCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the K-Nearest Neighbors (KNN) algorithm are TRUE?

- **A.** KNN is an instance-based 'lazy learning' algorithm that performs no explicit training phase
- **B.** The decision boundary is determined locally by the majority class among the k closest neighbors
- **C.** The choice of distance metric directly impacts classification results
- **D.** KNN produces a compact mathematical parametric formula independent of the training dataset

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the Softmax activation function in multi-class classification are TRUE?

- **A.** It transforms unnormalized model logits into a valid probability distribution
- **B.** The sum of all output probabilities across all mutually exclusive classes equals exactly 1
- **C.** Every individual output probability value is strictly bounded between 0 and 1
- **D.** Larger input logit values correspond monotonically to higher output probability values

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, the core method called on an estimator instance to train the machine learning algorithm on feature matrix X and target labels y is model.____(X, y).

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, after training an estimator, the method used to generate predicted target values for new unseen samples is model.____(X_test).

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn classification models (such as LogisticRegression), the method that returns calibrated class probability estimates for each sample is model.____(X_test).

Answer: _____

Page Answers: **Q91**: ABC • **Q92**: ABCD • **Q93**: fit • **Q94**: predict • **Q95**: predict_proba

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn preprocessing (e.g., StandardScaler), the combined method that calculates statistics on training data and applies scaling in a single step is scaler.____(X_train).

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn, the function from sklearn.model_selection commonly used to randomly partition a dataset into training and testing subsets is ____.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn's KMeans clustering estimator, the hyperparameter that specifies the desired number of clusters to form is ____.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In scikit-learn's LinearRegression, the learned weight slope coefficients of the model are stored in the model.____ attribute.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In NumPy, the function used to initialize an array of a given shape populated entirely with zeros is np.____().

Answer: _____

Page Answers: **Q96**: fit_transform • **Q97**: train_test_split • **Q98**: n_clusters • **Q99**: coef_ • **Q100**: zeros